Project 1: What Does Your Vision Model See?

Shuyan Zhang
Johns Hopkins University
szhan312@jhu.edu

Adriana Romero
Johns Hopkins University
aromer30@jhu.edu

Abstract

This project compares several image classification approaches, including linear classifiers, hand-crafted features, multilayer perceptrons (MLPs), convolutional neural networks (CNNs), and Vision Transformers (ViTs). In Part A, the models were trained and evaluated on the PASCAL VOC dataset, with model selection based on validation performance. The fine-tuned pretrained Vision Transformer (A6-H) achieved the highest test mean per-class accuracy of 92.1%. In Part B, the selected models were evaluated on photographs collected under ordinary and challenging conditions, including rotation, changes in distance, and occlusion. A6-H achieved 88% accuracy on ordinary photographs, while a targeted repair improved its accuracy on the 90-degree rotation subset from 40% to 90%. These findings demonstrate the advantages of pretrained representations and fine-tuning, while highlighting the importance of robustness evaluation.

1. Setup

We use object crops from PASCAL VOC 2012 [1]. We select five classes: sofa, bottle, chair, dining table, and bus, because they have different shapes, sizes, and visual features. The dataset contains 19 classes in total. Using the image-level split with seed 0, the dataset is divided into 7,541 training, 797 validation, and 8,381 test crops.

The images are first resized depending on the model: 32×32 pixels for the linear classifier and MLP, 64×64 for the hand-crafted features, CNN, and ViT trained from scratch, and 224×224 for the pretrained ViT. Data are standardized using the mean and standard deviation from the training crops. The images are flattened for the linear classifier and MLP, while the other models use their corresponding image representations. The image data are then converted into PyTorch tensors for further analysis and model training.

The linear classifier was trained using Adam with a learning rate of 0.001, a batch size of 256, and 60 epochs. Different L2 values (0.01, 1, and 10) were tested to evaluate their effect on performance and accuracy. The MLP, CNN, and ViT models were trained using AdamW with different learning rates and batch sizes depending on the model. The pretrained ViT was fine-tuned using a learning rate of 0.00001 for the encoder and 0.001 for the classification head. Model performance was evaluated using mean per-class accuracy, and validation accuracy was used for checkpoint selection where implemented. Training was performed using PyTorch with GPU acceleration.

2. Part A: a ladder of classifiers

Predictions. We expect A6 achieve the highest accuracy because it was pretrained on ImageNet-21k and the fine-tuned on imageNet-1k, which means it already learned useful information. We expect A4 achieve the second-highest accuracy. Its CNN uses convolutional layers to learn local image patterns. Early CNN detect simple features such as edges, lines and color, while deeper layers combine them into more complex features. A4 has 7 * 7 layers, but since it trained from scratch, it may not have the advantage of A6 has. We expect A5 have the lowest accuracy. Although it uses transformer attention to learn relationship, it is also trained from scratch and its 64 image patches may limit the amount of image it can capture, we placed it at the bottom. We expect A6 has an accuracy range from 70-80% and A4 has an accuracy range from 40-50% and A5 ranges from 30-40% . Our prediction is correct, A6 achieves an accuracy of 88% and A4 achieves a 56.2% with A5 getting a 35.3% for validation.

Results. The results in Table 1 show the performance of all evaluated models across the 19 classes and the five selected categories: sofa, bottle, chair, diningtable and bus. The input sizes and parameters, training time and hardware is also recorded for comparison between different models. The model ladder shows the gradual progression from traditional machine learning to deep learning approaches. Starting with a linear classifier (A1), A2 introduces hand-crafted features, followed by an MLP (A3), CNN (A4), and Vision Transformer (A5). Finally, A6 uses a pretrained Vision Transformer to leverage previously learned visual features.

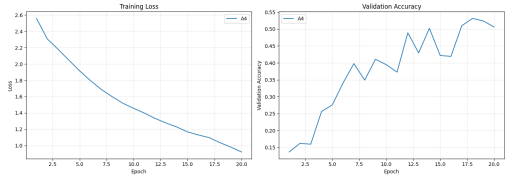


Figure 1. Training loss and validation accuracy of A4.

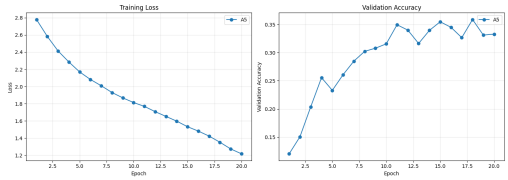


Figure 2. Training loss and validation accuracy of A5.

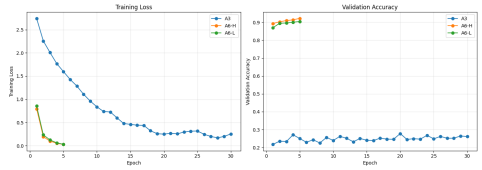


Figure 3. Training loss and validation accuracy of A3 and A6

Overall, pretrained A6 models achieved the highest classification accuracy with longer processing time.

Fair comparisons and diagnosis. Among the traditional approaches (A1 and A2) , concatenation achieved the highest mean per-class accuracy of 0.4375, followed by HOG at 0.3929. For models trained from scratch, A4 (CNN) achieved the best performance with an accuracy of 0.5246, outperforming both A3 (MLP, 0.2382) and A5 (ViT, 0.3453). A4 also achieved its highest accuracy on chair (0.7081), while its lowest accuracy was on sofa (0.3684). In contrast, the pretrained Vision Transformer models achieved substantially higher accuracy, with A6-H reaching 0.9209 and A6-L reaching 0.9057. Both models performed consistently well across the five selected classes, with bottle achieving the highest accuracy (0.9637 for A6-H and 0.9300 for A6-L). However, some classification errors occurred in A1 and A2 (Gabor). In A1, the dining table class had an accuracy of 0, while in A2, both sofa and bottle had an accuracy of 0. Since the other models performed successfully on these classes, these errors are unlikely to be caused by limitations in the dataset. Further investigation is needed to understand the cause. Regarding model complexity, A6 used 5,528,083 parameters and an input resolution of 224×224, compared with the smaller models trained from scratch. Its reported training time was 150.9 on a T4 GPU, while A6-L required 146.2. For the curve, as shown in Figures 1–3, the training loss for all model shows a decreasing

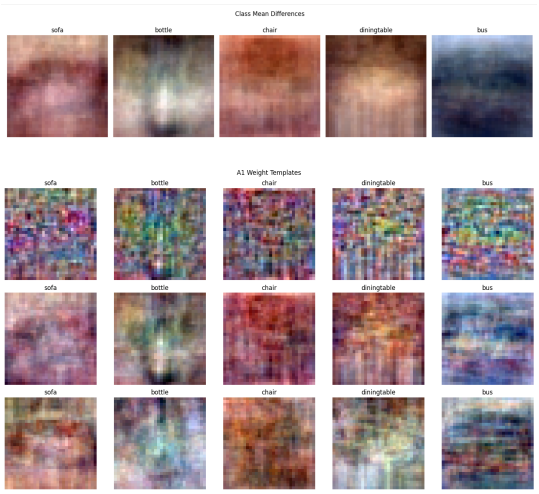


Figure 4. V1: Class mean differences (top) and A1 weight templates (bottom) for the five classes at three L2 settings. The VOC test mean per-class accuracy was 22.01% for L2 = 0.01, 17.06% for L2 = 1, and 12.90% for L2 = 10.

trend, while validation accuracy shows an increasing trend. The training loss for A6 is much shorter compare to A3, A4 and A5, their accuracy is much higher even with limited epochs. Among A3,A4 and A5, A3's accuracy rate increase much slower than A4 and A5.

Looking inside (V1–V5). Figures 4 to 7 show how different models learn from the images and how their features are represented. In V1, the class mean differences show the average visual patterns of the five classes, while the A1 weight templates show how the learned weights change under different L2 regularization settings. In V2, the Gabor filters capture different textures and orientations, while HOG focuses on local edges and their directions. These handcrafted features can capture some useful information, but they cannot perserve all the details. In V3, the eight selected MLP units look really noisy, it seems like a bunch of color patches, and it is difficult to identify clear shapes or specific patterns. This suggests that the MLP learns more general pixel-level patterns rather than easily recognizable object features. In V4, we examine the CNN kernels and feature maps from different stages to understand what the network learns as the features become deeper. Finally, V5 compares the patch-embedding components and positional embeddings of A5 and A6-H, showing how the two ViT models represent image patches and their spatial relationships.

	model	mean per-class (19)	accuracy on our five classes					input	params / dims	training data	time, hardware
			sofa	bottle	chair	diningtable	bus				
A1	linear, pixels	0.170627	0.077193	0.051813	0.512351	0	0.093190	32^2	58387	VOC	NaN, T4
A2	HSV histogram	0.14499	0.007018	0.012953	0.518753	0.003436	0	64^2	931	VOC	5.6,T4
	HOG	0.3929	0.196491	0.38342	0.460201	0.395189	0.483871	64^2	33535	VOC	7.6,T4
	Gabor	0.203535	0	0	0.661482	0	0.18638	64^2	323	VOC	5.4,T4
	concatenation	0.437494	0.221053	0.393782	0.509607	0.453608	0.512545	64^2	34751	VOC	5.4,T4
A3	MLP	0.2382	0.0982	0.15544	0.317475	0.109966	0.258065	32^2	791571	VOC	NaN,T4
A4	CNN, scratch	0.524564	0.3684	0.487	0.7081	0.5498	0.3763	64^2	701619	VOC	NaN,T4
A5	ViT, scratch	0.3453	0.1157	0.1865	0.4464	0.0996	0.344	64^2	828947	VOC	NaN,T4
A6-H	ViT-T/16, pretrained	0.9209	0.7438	0.9637	0.8856	0.7697	0.9498	224^2	5528083	IN-21k/1k + VOC	150.9,T4
A6-L	ViT-T/16, pretrained	0.9057	0.7333	0.93	0.881	0.7697	0.9319	$64^2 \rightarrow 224^2$	5528083	IN-21k/1k + VOC	146.2,T4

Table 1. Test results on the VOC 2012 `val` crops.

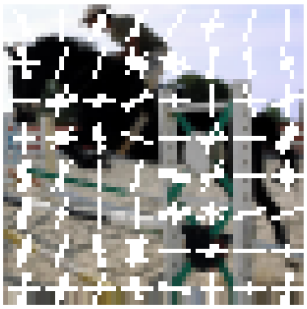


Figure 5. V2: the Gabor bank and HOG cells on one crop

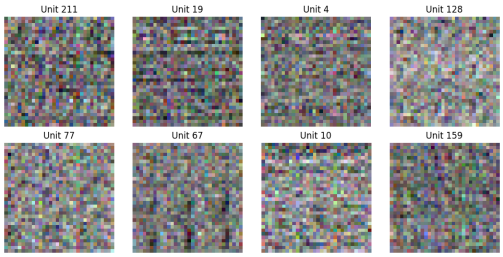


Figure 6. V3: eight MLP units

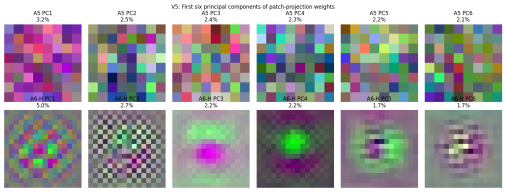


Figure 7. V5: for A5 and A6-H, patch-embedding components and position-embedding similarity.

3. Part B: testing the models in our world

Baseline (B1). We tested A2 (concatenated features), A4 (CNN), and A6-H (fine-tuned ViT) on 50 ordinary photographs, with 10 images from each of our five classes: bottle, bus, chair, dining table, and sofa. We wanted to see whether models evaluated on VOC crops would work equally well on pictures taken in our own surroundings. On these photographs, A2 achieved 20% mean per-class accuracy, compared with 41.81% on the same five VOC classes. A4 achieved 26%, compared with 40.44% on VOC. A6-H performed much better, reaching 88% on our photographs versus 86.26% on the five-class VOC evaluation.

Our photographs were taken in everyday settings rather than as standardized object crops. The objects could appear at different distances, against more distracting backgrounds, or under different lighting. These differences may help explain why A2 and A4 lost 21.81 and 14.44 percentage points, respectively. A6-H did not show the same drop on this small set: it improved by 1.74 percentage points. This does not mean our photographs were generally easier, but it does show that A6-H handled these particular images better than the other two models.

Predictions (B2). *Original milestone predictions unavailable in the provided notebook.* In B3, A6-H reached only 40% accuracy at 90-degree rotation, making this a particularly difficult condition in our experiment. The controlled photographs also show failures with occlusion and distance. For example, the same sofa was classified correctly when visible but confused with a chair when half covered, and the same bottle was recognized up close but not from farther away. These are observations from B3, not replacements for the predictions made before the experiment.

Controlled series (B3). We used 10 objects, two from each of our five classes, and photographed each object under nine conditions. For rotation, we used 0, 30, 90, and 180 degrees. For occlusion, we used no occlusion, quarter occlusion, and half occlusion. Our third factor was distance, with near and far photographs. This gave us 90 photographs in total, or 10 per condition. We ran A2, A4, and A6-H on the same images so that the comparisons were based on the same objects and changes.

Figure 9 shows the mean per-class accuracy for each condition. A6-H decreased from 100% at 0 degrees to 60% at 30 degrees and 40% at both 90 and 180 degrees. With occlusion, it dropped from 100% without coverage to 70% at quarter occlusion and 60% at half occlusion. For distance,

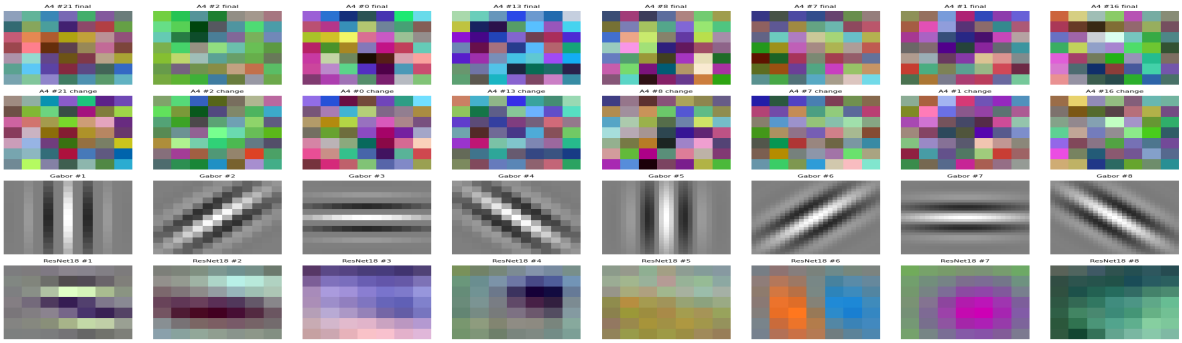


Figure 8. V4: (a) first-layer kernels of A4 (after training, and their change from initialization), the Gabor bank, and ResNet-18; (b) three channels per stage on one of our photos; (c) the four training crops that activate each channel most.

accuracy decreased from 100% near to 50% far. A2 and A4 stayed between 0% and 30% across these conditions. The individual examples also show why looking only at overall accuracy can hide important mistakes. For instance, A6-H recognized a bottle when it was near the camera but predicted dining table for the far photograph. It also recognized an unobstructed sofa but predicted chair after half of it was covered. These examples suggest that orientation, visible object parts, and apparent size can matter for recognition, although they do not prove which image features caused each error.

The series was useful because only one factor was deliberately changed at a time. Still, our sample included just two objects per class. The curves therefore describe these 10 objects and should not be interpreted as a general estimate of robustness for all bottles, buses, chairs, dining tables, or sofas.

Minimal pairs (B4). We selected three pairs where A6-H was correct in the first photograph but incorrect after changing one condition. The pairs use three different objects and show changes in distance, occlusion, and rotation.

In the first pair, a bottle was photographed near the camera and then farther away. For the near image, A6-H predicted bottle (1.000), dining table (0.000), and TV monitor (0.000). For the far image, its top predictions changed to dining table (0.978), sofa (0.011), and chair (0.006). A2 and A4 were wrong on both images. The main visible change was that the bottle appeared smaller in the far photograph. More background was also visible, which could have influenced the result, but the predictions alone cannot confirm the reason.

The second pair used a sofa, first with no occlusion and then with half of it covered. A6-H initially predicted sofa (0.997), chair (0.003), and potted plant (0.000). With half occlusion, the predictions became chair (0.994), sofa (0.003), and potted plant (0.002). A2 and A4 also failed on both images. The change suggests that hiding part of the sofa made it harder to distinguish from a chair, possibly because some useful shape information was missing.

The third pair used a bus at 0 and 90 degrees. At 0 degrees, A6-H predicted bus (0.999), train (0.001), and car (0.000). At 90 degrees, it predicted train (0.585), motorbike (0.332), and bottle (0.031). Again, A2 and A4 were incorrect on both images. The object had not changed, only its orientation. This illustrates how a rotated appearance can produce a different prediction even when the original view was recognized with high confidence. The three pairs are shown in Figures 10, 11, and 12.

One repair (B5). We focused the repair on 90-degree rotation, where the original A6-H model achieved only 40% accuracy in B3. The notebook loads the original A6-H checkpoint and a separate repaired checkpoint, then evaluates both on the same image sets. The assignment specifies 10 adaptation photographs (two per class) and a default procedure of 100 updates, with eight VOC training crops and eight adaptation photographs per update. However, the uploaded B5 code loads an already repaired model rather than showing the optimization run. We therefore cannot verify that those exact training settings or stopping rules were used for the saved checkpoint.

The repaired model performed much better on the targeted condition: accuracy at 90-degree rotation rose from 40% to 90%, or from four correct photographs to nine out of ten. Across the 70 unique B3 images, accuracy increased from 60% to 80%. This matters because the improvement was not limited to the ten rotation photographs. On the five held-out challenging photographs, accuracy increased from 20% to 60%, while accuracy on the five held-out ordinary photographs stayed at 80%.

We also checked whether the repair changed performance on the original B1 photographs and the VOC test set. On B1, accuracy increased from 88% to 92%. On the full 19-class VOC test set, however, mean per-class accuracy decreased slightly from 92.10% to 92.00%. The re-

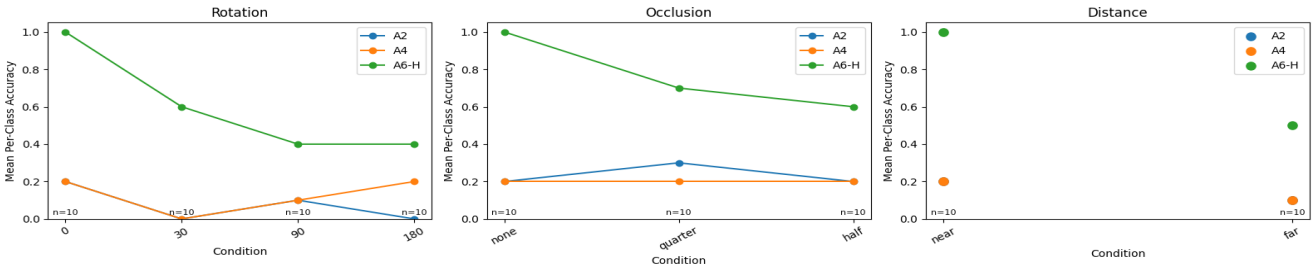


Figure 9. B3: mean per-class accuracy over our five classes for each condition of rotation, occlusion, and distance, for A2, A4, and A6-H. Each plotted condition contains 10 photographs, two per class.

object	rotation				occlusion			distance	
	0	30	90	180	none	quarter	half	near	far
bottle/object1	0/0/1	0/0/0	0/0/1	0/0/1	0/0/1	0/0/0	0/0/0	0/0/1	0/0/0
bottle/object2	0/0/1	0/0/1	0/0/1	0/0/1	0/0/1	1/0/1	0/0/1	0/0/1	0/0/0
bus/object1	0/0/1	0/0/0	0/0/0	0/0/0	0/0/1	0/0/0	0/0/0	0/0/1	0/0/1
bus/object2	0/0/1	0/0/1	0/0/0	0/0/0	0/0/1	0/0/1	0/0/1	0/0/1	0/0/1
chair/object1	1/1/1	0/0/1	1/1/1	0/1/1	1/1/1	1/1/1	1/1/1	1/1/1	1/0/0
chair/object2	1/0/1	0/0/1	0/0/1	0/1/1	1/0/1	1/1/1	1/1/1	1/0/1	0/0/0
diningtable/object1	0/1/1	0/0/1	0/0/0	0/0/0	0/1/1	0/0/1	0/0/1	0/1/1	0/1/1
diningtable/object2	0/0/1	0/0/0	0/0/0	0/0/0	0/0/1	0/0/1	0/0/1	0/0/1	0/0/1
sofa/object1	0/0/1	0/0/0	0/0/0	0/0/0	0/0/1	0/0/1	0/0/0	0/0/1	0/0/1
sofa/object2	0/0/1	0/0/1	0/0/0	0/0/0	0/0/1	0/0/0	0/0/0	0/0/1	0/0/0

Table 2. B3 per object: each cell is intended to show whether A2/A4/A6-H were correct (1) or incorrect (0), in that order. All values are taken from the exported `b3_object_results.csv` file.





Figure 10. Pair 1: The same bottle photographed near and far. A6-H recognizes the near bottle but predicts dining table when it is farther away.





Figure 11. Pair 2: The same sofa without occlusion and with half occlusion. A6-H changes its prediction from sofa to chair.

pair therefore helped on the selected rotation condition and several other photograph evaluations, but did not improve every result. The VOC change was small, so it would be too strong to conclude that the repair caused a major loss in general performance.

4. Frozen features or fine-tuning? (665 only)

We compared a frozen pretrained ViT with A6-H to investigate the effect of fine-tuning the feature representation. For the frozen approach, we loaded the original ImageNet-pretrained ViT-Tiny model and froze all other encoder parameters. We extracted important features from both the original and horizontally flipped training images,





Figure 12. Pair 3: The same bus at 0 and 90 degrees. A6-H changes its prediction from bus to train after rotation.

evaluation set	n	original A6-H	repaired
B3, rotation 90	10	40%	90%
B3, all images (each once)	70	60%	80%
held-out, ordinary	5	80%	80%
held-out, challenging	5	20%	60%
B1	50	88%	92%
VOC test (19 classes)	8381	92.10%	92.00%

Table 3. B5: mean per-class accuracy before and after the repair targeting 90-degree rotation.

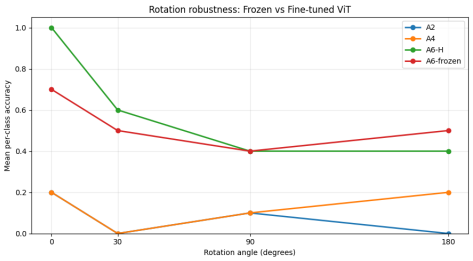


Figure 13. Rotation for Frozen vs A6-H

model	VOC (19)	B1	total params	trainable	cost
frozen + linear head	0.8883	0.84	5528083	3667	extract + fit
A6-H, fine-tuned	0.9209	0.88	5528083	5528083	all params are trained

Table 4. Frozen features vs. fine-tuning, mean per-class accuracy.

then trained a new linear classification head using AdamW. We use a learning rate of 1e-3, weight decay of 1e-2, batch size of 32, and 5 epochs. During each epoch, the original or flipped feature was randomly selected for each training image. We use the validation accuracy to choose the best model, which is epoch 4 with a accuracy of 0.8793. We then compared the frozen model with A6-H using B1 photos and B3 rotation images. We also compared their parameter counts and training costs to understand the trade-offs between using frozen pretrained features and fine-tuning the entire model.

According to Figure 13, A6-H and A6-frozen show better rotation performance than A2 and A4, which is expected because at 0 degrees, when there is no rotation, the overall performance of A6 is much better than the other models. A6-H achieves an accuracy of 1.0, while A6-frozen achieves 0.7. As rotation increases, the accuracy of both A6 models starts to decrease, reaching around 0.4 at 90 degrees. However, A2 and A4 show a slight increase in accuracy at 90 degrees compared with their results at 30 degrees. When the rotation reaches 180 degrees, A6-frozen and A4 show a slight increase in accuracy, while A6-H remains the same and A2 decreases to 0. The rotation plot match the table 4, as A6-H and A6 frozen has a pretty high accuracy rate of VOC and B1, but since A6 trained all parameters, it improve the accuracy when training normal or slightly rotated images. A6-frozen retains the original ImageNet representation, these features may preserve more general visual information, which could be the reason it outperform A6-H during large rotation.

5. Conclusion

In this project, we compared different image classification models, from linear classifiers and hand-crafted features to CNNs and Vision Transformers. A6-H achieved the best overall performance on VOC, reaching 92.09% mean per-class accuracy, and also performed well on our ordinary photographs with 88% accuracy. However, the controlled experiments showed that its predictions could change significantly with rotation, occlusion, and distance. The repair experiment improved accuracy on 90-degree rotation from 40% to 90%, although VOC accuracy decreased slightly. We also found that fine-tuning provided better accuracy than frozen features, but required training many more parameters. Overall, the experiments showed that strong performance on a standard dataset does not guarantee equally reliable predictions under different real-world conditions.

References

[1] Mark Everingham, Luc Van Gool, Christopher K. I. Williams, John Winn, and Andrew Zisserman. The PASCAL visual object classes (VOC) challenge. *International Journal of Computer Vision*, 88(2):303–338, 2010.

Contributions

Shuyan: Wrote the code for A1, A3–A5 and V1-V4 and a few code in the extension 665. Took all the pictures and organized into folders. Finish Part A and 665 section of the report. Adriana: wrote the code for A2, A6, and most of Part B, as well as some code for the extension 665. She also helped organize and evaluate the experimental results, created the tables and figures, and completed the Setup, .